



2 (a) Define specific heat capacity.

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..... [2]

(b) Two solid blocks X and Y are made from different metals. The blocks have different initial temperatures. Block Y is initially at room temperature.

The blocks are placed in direct thermal contact with each other at time $t = 0$. Fig. 2.1 shows the variation with t of the temperatures of the two blocks.

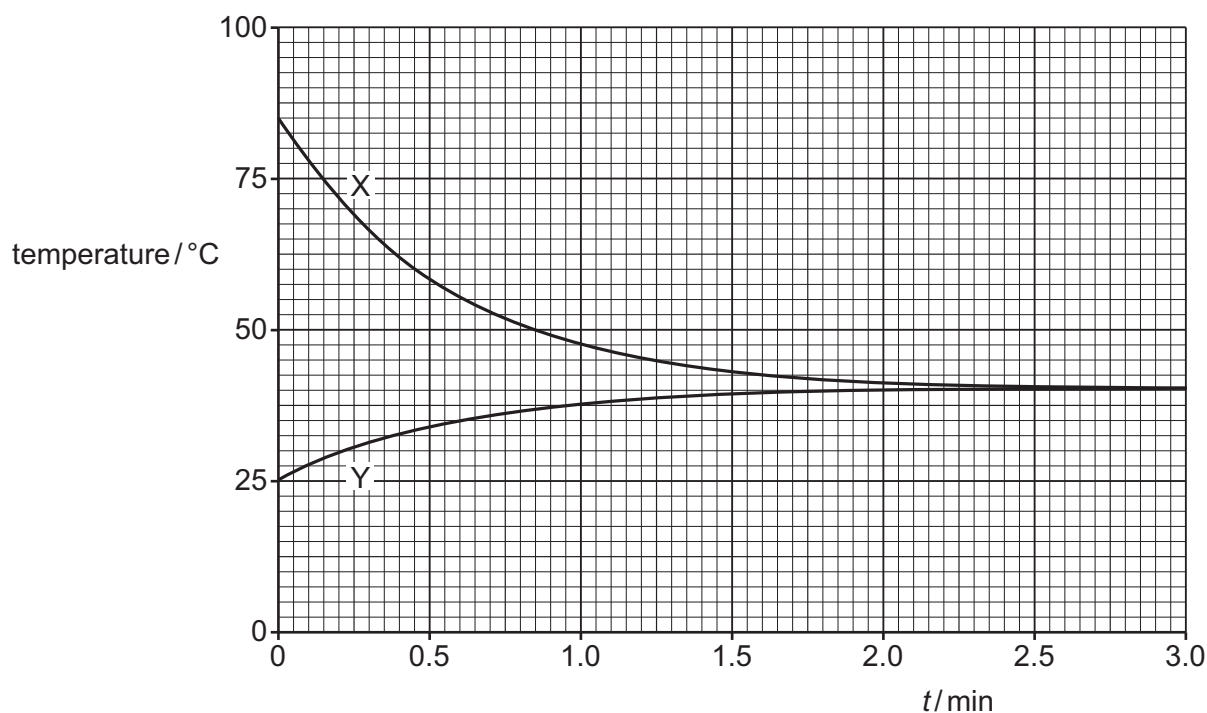


Fig. 2.1





- (i) State **three** conclusions that may be drawn from Fig. 2.1. The conclusions may be qualitative or quantitative.

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[3]

- (ii) The ratio $\frac{\text{mass of block Y}}{\text{mass of block X}}$ is equal to 1.3.

The metal in block Y has a specific heat capacity of $901 \text{ J kg}^{-1} \text{ K}^{-1}$.

Determine the specific heat capacity of the metal in block X.

specific heat capacity = $\text{J kg}^{-1} \text{ K}^{-1}$ [3]

[Total: 8]

